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ACCESSORY SYSTEM FOR A VEHICLE HAVING A PICKUP BED

Abstract

An accessory system for a vehicle having a pickup bed, including a frame configured for selectively connecting to a rear portion of the vehicle, the frame being connected to and extending over the pickup bed when the frame is installed on the vehicle. The frame is arranged and configured to selectively receive a plurality of cooperating accessories, each accessory of the plurality of cooperating accessories being configured to connect to the frame individually and simultaneously with other accessories of the plurality of cooperating accessories. A toolbox for an accessory system including a toolbox body defining a cavity and shaped and arranged to be received in an accessory system frame; and a toolbox door including: a latch defined in a bottom side of the toolbox door, and a handle for selectively opening the latch, the handle being disposed on the bottom side of the toolbox door.

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Background/Summary

CROSS-REFERENCE [0001] The present application claims priority to U.S. Provisional Patent Application No. 63/560,388, entitled “Accessory System for a Vehicle having a Pickup Bed,” filed Mar. 1, 2024, the entirety of which is incorporated by reference herein.

TECHNICAL FIELD

[0002] The present technology relates to accessory systems for vehicles having a pickup bed.

BACKGROUND

[0003] Vehicles with pickup beds, generally referred to as pickup trucks, are commonly used for labor, either professional or recreational. To supplement or customize transport capabilities, these vehicles sometimes have retrofits to transport tools, supplies, etc. In some cases, the retrofit could include additional structures or accessories installed in or over the pickup bed.

[0004] Different types of work, or even different tasks in the same line of work, may require different vehicle retrofits. For instance, toolboxes to carry tools may be required for one task, while a canopy to protect objects in the pickup bed may required for another. These different needs may be conflicting, requiring the removal of one structure to permit the installation of another. In some cases, some structures such as canopies are not easily removed and may leave holes or damage where the structure was attached to the vehicle body.

[0005] In some cases, multiple retrofit structures or accessories may be installed together by modification thereof. For example, ladder racks may be added to a canopy by drilling holes in the canopy to fasten the ladder racks thereto. After-market addition of the ladder racks is thus not convenient, especially for an unplanned need. Further, selective removal of the ladder racks would leave holes to be repaired in such an instance.

[0006] Therefore, there is a desire for a solution addressing at least some of these drawbacks.

SUMMARY

[0007] It is an object of the present technology to ameliorate at least some of the inconveniences present in the prior art.

[0008] In accordance with an aspect of the present technology, there is provided an accessory system to be connected to the vehicle and extending over the pickup bed. The system includes a frame that can be selectively connected to or removed from the pickup bed, permitting the user install or remove the system as required. The frame is connected to the vehicle using clamps by the present technology, permitting a drill-less installation of the system on the vehicle. The rigid members of the frame are further configured to fasten together, such that the frame construction (either on the vehicle directly or before connecting to the vehicle) also does not require drilling, welding, and/or modification of the frame or frame members.

[0009] By the present technology, the accessory system is further configured to permit the installation of one or many of the cooperating accessories simultaneously. The cooperating accessories include, but are not limited to, toolboxes, one or more top rack members, a cab rack extension, a cargo rack, a canopy, a rear door, one or more ladder racks, one or more mounting hooks, and a cab barrier. For instance and as is illustrated herein, the frame is configured and arranged to simultaneously receive thereon: two toolboxes, a cargo rack formed from three top rack members, a cab rack extension, a canopy, a rear door, and two ladder racks. The frame is also configured to be installed on the vehicle with fewer or none of these accessories. In some cases,

one or more side doors could be included in place of a toolbox or toolboxes. Each of the accessories is installed using fasteners and slots defined in the frame, thereby providing drill-less installation of the accessories in the frame as well.

[0010] By the present technology, toolboxes for selectively connecting to the frame and for selective use on a vehicle with a pickup bed are also disclosed. The toolbox has a toolbox body defining a cavity configured and shaped to receive objects therein and a toolbox door pivotally connected to the toolbox body. The toolbox door selectively closes the cavity and opens to an exterior of the frame, such that the contents of the toolbox can be accessed from an exterior of the vehicle. The latch and handle of the toolbox door are disposed on a bottom edge thereof, leaving a maximum expanse of the exterior side of the toolbox door generally uniform. This permits application of markings, logos, advertisements, and the like. The toolbox body closes a lateral side opening of the frame, and extends into a space interior to the frame.

[0011] In accordance with an aspect of the present technology, there is provided an accessory system for a vehicle having a pickup bed, the system including a frame configured for selectively connecting to a rear portion of the vehicle, the frame being connected to and extending over the pickup bed when the frame is installed on the vehicle, the frame being arranged and configured to selectively receive a plurality of cooperating accessories, each accessory of the plurality of cooperating accessories being configured to connect to the frame individually and simultaneously with other accessories of the plurality of cooperating accessories.

[0012] In some embodiments, the accessory system further includes at least one accessory of the plurality of cooperating accessories.

[0013] In some embodiments, the plurality of cooperating accessories comprises at least one of: at least one toolbox; at least one top rack member; a cargo rack formed from a plurality of rack rigid members; a cab rack extension; a canopy; a rear door; a ladder rack; at least one mounting hook; at least one side door; and a cab barrier.

[0014] In some embodiments, the at least one toolbox further includes a toolbox body defining a cavity configured and shaped to receive objects therein, and a toolbox door pivotally connected to the toolbox body, the toolbox door selectively closing the cavity; and when the at least one toolbox is selectively connected to the frame: the toolbox body closes a lateral side opening of the frame, and the toolbox door opens to an exterior of the frame.

[0015] In some embodiments, the accessory system further includes a plurality of clamps for selectively connecting the frame to side walls of the pickup bed.

[0016] In some embodiments, the frame includes a first base frame member configured to selectively connect to a first side wall of the pickup bed; a second base frame member configured to selectively connect to a second side wall of the pickup bed; two first rising frame members connected to the first base frame member, a lower end portion of each first rising frame member being connected to the first base frame member; two second rising frame members connected to the second base frame member, a lower end portion of each second rising frame member being connected to the first base frame member; a first top frame member connected to upper end portions of the two first rising frame members; a second top frame member connected to upper end portions of the two second rising frame members; and a third top frame member extending between and connected to the first top frame member and the second top frame member.

[0017] In some embodiments, the first top frame member extends parallel to the first base frame member; and the second top frame member extends parallel to the second base frame member.

[0018] In some embodiments, each of the first rising frame members extends upward and inward from the first base frame member to the first top frame member; and each of the second rising frame members extends upward and inward from the second base frame member to the second top frame member.

[0019] In some embodiments, the third top frame member includes a rigid member; and at least one extension member moveably connected to the rigid member, the at least one extension member

being selectively slidable relative to the rigid member permitting an adjustment of a width of the frame.

[0020] In some embodiments, each of the first rising frame members and each of the second rising frame members has two slots defined therein, the two slots extending from the lower end portion to the upper end portion.

[0021] In some embodiments, the two slots of each of the first rising frame members and each of the second rising frame members is shaped to selectively secure at least one fastener therein, each slot having an opening narrower than a largest width of the slot.

[0022] In some embodiments, each of the first top frame member and the second top frame member includes a top frame member body having four sides extending from a rear of the frame to a front portion of the frame, the top frame member body having at least one slot defined in each of the four sides.

[0023] In some embodiments, each of the four slots of the first top frame member and the second rising frame member is shaped to selectively secure at least one fastener therein, each slot having an opening narrower than a maximum width of the slot.

[0024] In some embodiments, the first base frame member, the second base frame member, the two first rising frame members, the two second rising frame members, the first top frame member, the second top frame member, and the third top frame member are formed from extruded aluminum.

[0025] In some embodiments, each of the first base frame member and the second base frame member further comprises a plurality of clamps for selectively connecting the frame to side walls of the pickup bed.

[0026] In some embodiments, the first top frame member and the second top frame member each define a longitudinal slot therein; and the longitudinal slot is shaped to receive a portion of a canopy for selectively connecting the canopy to the frame.

[0027] In accordance with another aspect of the present technology, there is provided an accessory system frame for a vehicle having a pickup bed, the frame including a first base frame member configured to selectively connect to a first side wall of the pickup bed; a second base frame member configured to selectively connect to a second side wall of the pickup bed; two first rising frame members connected to the first base frame member, a lower end portion of each first rising member being connected to the first base frame member; two second rising frame members connected to the second base frame member, a lower end portion of each second rising member being connected to the first base frame member; a first top frame member connected to upper end portions of the two first rising frame members; a second top frame member connected to upper end portions of the two second rising frame members; and a third top frame member extending between and connected to the first top frame member and the second top frame member, the frame being configured to selectively connect to a rear portion of the vehicle, the frame being connected to and extending over the pickup bed when the frame is installed on the vehicle, the frame being arranged and configured to selectively receive a plurality of cooperating accessories, each accessory of the plurality of cooperating accessories being configured to connect to the frame individually and simultaneously with other accessories of the plurality of cooperating accessories.

[0028] In some embodiments, the first top frame member extends parallel to the first base frame member; and the second top frame member extends parallel to the second base frame member.

[0029] In some embodiments, each of the first rising frame members extends upward and inward from the first base frame member to the first top frame member; and each of the second rising frame members extends upward and inward from the second base frame member to the second top frame member.

[0030] In some embodiments, the third top frame member includes a rigid member; and at least one extension member moveably connected to the rigid member, the at least one extension member being selectively slidable relative to the rigid member permitting an adjustment of a width of the frame.

[0031] In some embodiments, each of the first rising frame members and each of the second rising frame members has two slots defined therein, the two slots extending from the lower end portion to the upper end portion.

[0032] In some embodiments, the two slots of each of the first rising frame members and each of the second rising frame members is shaped to selectively secure at least one fastener therein, each slot having an opening narrower than a largest width of the slot.

[0033] In some embodiments, each of the first top frame member and the second top frame member includes a top frame member body having four sides extending from a rear of the frame to a front portion of the frame, the top frame member body having at least one slot defined in each of the four sides.

[0034] In some embodiments, each of the four slots of the first top frame member and the second rising frame member is shaped to selectively secure at least one fastener therein, each slot having an opening narrower than a maximum width of the slot.

[0035] In some embodiments, the first base frame member, the second base frame member, the two first rising frame members, the two second rising frame members, the first top frame member, the second top frame member, and the third top frame member are formed from extruded aluminum.

[0036] In some embodiments, each of the first base frame member and the second base frame member further comprises a plurality of clamps for selectively connecting the frame to side walls of the pickup bed.

[0037] In some embodiments, the frame further includes a fourth top frame member extending between and connected to the first top frame member and the second top frame member, the fourth top frame member being disposed rearward of the third top frame member.

[0038] In some embodiments, the frame further includes a plurality of clamps for selectively connecting the first base frame member and the second base frame member to side walls of the pickup bed.

[0039] In accordance with yet another aspect of the present technology, there is provided a toolbox for an accessory system, the toolbox including a toolbox body defining a cavity configured and shaped to receive objects therein, the toolbox body being shaped and arranged to be received in an accessory system frame; and a toolbox door pivotally connected to the toolbox body, the toolbox door selectively closing the cavity, the toolbox door including: a latch defined in a bottom side of the toolbox door, and a handle for selectively opening the latch, the handle being disposed on the bottom side of the toolbox door.

[0040] In some embodiments, the toolbox door further comprises a sealing gasket disposed on an interior side of the toolbox door.

[0041] In some embodiments, the toolbox body forms a gutter arrangement.

[0042] In some embodiments, the toolbox further includes a composite end cap for selectively fastening objects to the toolbox body.

[0043] For purposes of this application, terms related to spatial orientation when referring to the accessory system or components thereof such as forwardly, rearwardly, top, bottom, left, and right are as they would normally be understood by a driver of the vehicle sitting thereon in a normal riding position with the accessory system installed thereon.

[0044] Embodiments of the present technology each have at least one of the above-mentioned aspects, but do not necessarily have all of them. Additional and/or alternative features, aspects, and advantages of embodiments of the present technology will become apparent from the following description, the accompanying drawings, and the appended claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0045] For a better understanding of the present technology, as well as other aspects and further features thereof, reference is made to the following description which is to be used in conjunction with the accompanying drawings, where:

[0046] FIG. **1** is a top, rear, right side perspective view of a vehicle with an accessory system according to non-limiting embodiments of the present technology mounted thereon;

[0047] FIG. **2** is a top, rear, right side perspective view of the accessory system of FIG. **1**;

[0048] FIG. **3** is another top, rear, right side perspective view of the accessory system of FIG. **1**;

[0049] FIG. **4** is a top plan view of the accessory system of FIG. **1**;

[0050] FIG. **5** is a cross-sectional view of the accessory system of FIG. **1**, taken along line 5-5 of FIG. **4**;

[0051] FIG. **6** is a close-up, cross-sectional view of portions of a frame of the accessory system of FIG. **1**, taken from portion **6** of FIG. **5**;

[0052] FIG. **7** is a cross-sectional view of the frame of the accessory system of FIG. **1**, taken along line 7-7 of FIG. **5**;

[0053] FIG. **8** is a top, front, left side perspective view of the accessory system of FIG. **1**, with additional accessories of the system connected thereto;

[0054] FIG. **9** is a rear elevation view of the accessory system of FIG. **8**;

[0055] FIG. **10** is a left side elevation view of the accessory system of FIG. **8**;

[0056] FIG. **11** is a front, left side perspective view of the accessory system of FIG. **8**;

[0057] FIG. **12** is a rear, right side perspective view of the accessory system of FIG. **8**;

[0058] FIG. **13** is a top, rear, right side perspective view of the vehicle with the accessory system mounted thereon, with an alternative set of accessories connected thereto;

[0059] FIG. **14** is a top, front, left side perspective view of the accessory system of FIG. **1**, with another alternative set accessories of the system connected thereto;

[0060] FIG. **15** is a rear, left side perspective view of the accessory system and accessories of FIG. **14**;

[0061] FIG. **16** is a rear, side perspective view of a toolbox of the accessory system of FIG. **1**; and

[0062] FIG. **17** is a bottom, side perspective view of the toolbox of FIG. **16**.

[0063] It should be noted that the Figures may not be drawn to scale, unless otherwise noted.

DETAILED DESCRIPTION

[0064] In accordance with the present technology, an accessory system **100** for a vehicle **10** having a pickup bed is described. Such vehicles, generally those referred to as pickup trucks, have a cab for the driver and one or more passengers and a pickup bed. The pickup bed, as generally described herein, is defined by a rear side of the cab, two side walls, and a rear side door (generally referred to as a tailgate).

[0065] With reference to FIG. **1**, the accessory system **100**, also referred to herein as the system **100**, is provided to selectively connect one or more cooperating accessories to the vehicle **10**. To receive the one or more accessories, the system **100** includes a frame **101** for selectively connecting to the vehicle **10**, specifically to a rear portion of the vehicle **10**. When the system **100** is connected to and installed on the vehicle **10**, the frame **101** extends over the pickup bed of the vehicle **10**. As will be described in more detail below, the frame **101** is selectively connected to the pickup bed side walls.

[0066] The frame **101** is arranged and configured to selectively receive a plurality of cooperating accessories. Specifics of the different accessories will be described in more detail below. By the present technology, each accessory is shaped and configured to connect to the frame **101** individually and simultaneously with other cooperating accessories of the system **100**. By the present technology, the system **100** includes a series of clamps **99** for selectively connecting the frame **101** to side walls of the pickup bed. It is contemplated that the frame **101** could be

selectively connected to the vehicle **10** in different manners.

[0067] With additional reference to FIGS. **2** to **5**, the frame **101** includes a left base frame member **110** configured to selectively connect to a left side wall of the pickup bed and a right base frame member **110** configured to selectively connect to a right side wall of the pickup bed. Each base member **110** is shaped to rest on an upper surface and along the longitudinal length of the corresponding side wall. Weight of the frame **101**, as well the weight of accessories and objects supported thereby, are thus transferred to the vehicle **10** via the base frame members **110**. The clamps **99** anchor the base frame members **110** to the corresponding side wall. In the illustrated embodiment, the clamps **99** are applied to the base frame members **110** when installing the frame **101** to the vehicle **10**. In some embodiments, one or more clamps **99** could be integrally formed with or affixed to the base frame members **110**.

[0068] The frame **101** further includes two rising frame members **130** connected to each base frame member **110**. A lower end portion of each rising member **130** is connected to the corresponding base frame member **110**. Specifically, the lower end portion of each rising member **130** is fastened to the corresponding base frame member **110**.

[0069] In the illustrated embodiment, one rising frame member **130** is positioned at a front end and a rear end of each base frame member **110**. There is thus one rising frame member **130** at each corner of the frame **101**. It is contemplated that the exact positioning of the rising frame members **130** could vary in some embodiments.

[0070] The frame **101** further includes a left top frame member **150** connected to upper end portions of the two left rising frame members **130** and a right top frame member **150** connected to upper end portions of the two right rising frame members **130**. A front (third) top frame member **160** extends between and is connected to the left and right top frame members **150**. A left end portion of the front top frame member **160** is fastened to a forward portion of the left top frame member **150** and a right end portion of the front top frame member **160** is fastened to a forward portion of the right top frame member **150**.

[0071] The left top frame member **150** extends parallel to the left base frame member **110** and the right top frame member **150** extends parallel to the right base frame member **110**. In the illustrated embodiment, the top members **150** and the base members **110** are further all arranged parallel to one another. Each of the rising frame members **130** extends upward and inward from its corresponding base frame member **110** to its corresponding top frame member **150**. As such, the frame **101** forms a trapezoidal volume, similar in shape to standard pickup truck canopies. The exact shape and volume formed by the frame **101** could vary in different embodiments.

[0072] With additional reference to FIG. **6**, the front top frame member **160** is formed from a rigid member **162** and one or more extension members **164** moveably connected to the rigid member **162**. In the illustrated embodiment the frame member **160** includes two extension members **164**. It is contemplated that the frame member **160** could have more or fewer extension members **164**, depending on the embodiment. Each extension member **164** is selectively slidable relative to the rigid member **162**, specifically along a slot **165** in the present embodiment. The sliding extension member **164** permits the frame member **160** to change in length, thereby adjusting of a width of the frame **101**. A fastener **167**, a bolt **167** as illustrated, extends through the slot **165** and through an aperture **163** defined in the rigid member **162** to selectively fix the relative position of the extension member **164** relative to the rigid member **162**. In the present embodiment, the rigid member **162** has three apertures **163** defined in each end portion, but the exact number could vary. The total length of the front top frame member **160**, and thus the width of the frame **101**, is thus selectively adjustable by loosening one or both fasteners **167** and moving one or both of the extension members **164** relative to the rigid member **162**. The adjustable width permits the user to adjust the width of the frame **101** to the exact width of the pickup bed, as vehicles of a same make and model may have small variability in their exact pickup bed width. For example, some vehicles **10** may have a liner or coatings applied to the walls of the pickup bed, which would slightly decrease the

separation between the pickup bed side walls. The adjustable width of the frame **101** may also permit a greater ease of installation and securing the frame **101** in place on the vehicle **10**. In some embodiments, the extension member **164** could have a plurality of laterally-spaced holes defined therein in place of the slot **165**. The plurality of laterally-spaced holes would be arranged to permit the fastener **167** to selectively adjust the length of the front top frame member **160** as described above. The embodiment of the extension member **164** having the plurality of laterally-spaced holes could be used with the rigid member **162** with the apertures **163** described above. It is also contemplated that the embodiment of the extension member **164** having the plurality of laterally-spaced holes could be used with a rigid member having a slot defined therein.

[0073] In the illustrated embodiment, the frame **101** further includes a rear (fourth) top frame member **221**. The member **221** extends between and is connected to the left and right top frame members **150**. Specifically, the top frame member **221** is disposed rearward of the front top frame member **160**. Depending on the aspect ratio of the frame members **110**, **130**, **150**, **160**, it is contemplated that the rear frame member could be omitted in some embodiments.

[0074] In some implementations, the frame **101** could include a rear (fourth) top frame member extending between and connected to the left and right top frame members **150**, similar to the front (third) top frame member **160**. Such a rear top frame member could be implemented alternatively or additionally to the top frame member **221**. A left end portion of such a rear top frame member would be fastened to a rear portion of the left top frame member **150** and a right end portion of the rear top frame member would be fastened to a rear portion of the right top frame member **150**.

[0075] In the illustrated embodiment, the rear frame member **221** is formed by a rigid member **221** that simultaneously serves as a rack member for a roof rack **225** for the system **100**. The rigid member **221** is fastened to a top side of the left and right top members **150**, and extends thereabove. It is contemplated that the frame **101** could include a rear frame member aligned and parallel to the front top frame member **160** in some embodiments.

[0076] With continued reference to FIG. **6**, each of the left and right top frame members **150** has a top frame member body **152** having four sides extending from a rear of the frame **101** to a front portion of the frame **101**. Each top frame member body **152** has a slot **155** defined in each of the four sides. Each slot **155** is shaped to selectively secure one or more fasteners therein, each slot **155** having an opening narrower than a maximum width of the slot **155**. In this way, the head of a fastener (such as a bolt) can slide into the slot **155** at a front or rear portion of the member **150**, and then an item can subsequently be fastened to the fastener, in turn selectively securing the item to the frame **101**. As is illustrated, the front top member **160** and the rising member **130** are fastened to the top member **150** via the slots **155**. In the present embodiment, the rack members **220** are also fastened to the member **150** via the slots **155**.

[0077] In some embodiments, each of the left and right top frame members **150** could further have a longitudinal slot formed in the top frame member body **152**. The longitudinal slot is shaped to receive a portion of a canopy **250** for selectively connecting the canopy **250** to the frame **101**; see for instance FIGS. **13** to **15**. By providing a selective attachment of the canopy **250** to the frame **101**, the system **100** may be easily converted from being open on a top side to closed. For instance, the canopy **250** could be removed to carry large items or equipment and replaced for inclement weather.

[0078] With additional reference to FIG. **7**, each of the rising frame members **130** has two slots **134** defined therein. The two slots **134** extend from the lower end portion of each rising frame member **130** to the upper end portion thereof. The slots **134** are defined in an exterior facing side of the frame members **130**. It is contemplated that additional sides of the members **130** could have similar slots defined therein. It is also contemplated that one or more rising frame members **130** could have more or fewer than two slots defined therein.

[0079] Similarly to the slots **155**, each slot **134** is shaped to selectively secure fasteners therein. Each slot **134** has an opening narrower than a largest width of the slot **134**. In this way, the head of

a fastener (such as a bolt) can slide into the slot **134** at a top portion of the member **130**, and then an item can then be fastened to the fastener, in turn selectively securing the item to the frame **101**. [0080] The base frame members **110**, the rising frame members **130**, and the top frame members **150** are formed from extruded aluminum. It is contemplated that different surface treatments and processing could be applied thereto, specifics of which depend on the embodiment. By forming the members **110**, **130**, **150** by extrusion, including directly forming the slots **134** and the slots **155**, the frame **101** can be sized to a variety of different vehicles **10** without requiring complicated design changes. The members **110**, **130**, **140** are simply sized to the pickup bed upon formation. It is contemplated that different materials and fabrication methods could be implemented in different embodiments.

[0081] As is mentioned briefly above, the members **110**, **130**, **140** are all fastened together. As such, welding is not required: neither prior to shipping to an end user (thereby facilitating shipping) nor by the final user upon installation on the vehicle **10** (thereby simplifying installation).

[0082] As is noted above, the system **100** is configured and arranged to provide selective connection of multiple different accessories to the vehicle **10**, over the pickup bed. By the present technology, the frame **101** is shaped and arranged to receive multiple of the following accessories simultaneously: one or more toolboxes **200**; one or more top rack members **220**; a cab rack extension **230**; a canopy **250**; a rear canopy door **260**; one or more ladder racks **270**; one or more side doors **280**; one or more mounting hooks (not shown); and a cab barrier (not shown). The cab rack extension **230** connects to the frame **101** and extends forward over the cab of the vehicle **10**. The cab barrier is a separator disposed between the frame **101** and the cab of the vehicle **10**. In at least some cases, the cab barrier can aid in protecting the cab from movement of items inside the pickup bed. In some embodiments, the system **100** includes a cargo rack **225** formed from two or more rack rigid members **220**, **221**. See, for instance, FIG. **1**. It is also contemplated that an integrally formed cargo rack could be provided in some cases.

[0083] In some embodiments the system **100** could be provided with only the frame **101** for installation on the vehicle **10**, with accessories to be procured separately. In some embodiments, the system **100** could be provided with the frame **101** and one or more of the above described accessories.

[0084] It further noted that system **100** is configured to receive any and all of the above noted accessories simultaneously, except where the two accessories are meant to occupy the same space. For example, the toolboxes **200** and the side doors **280** are both configured to occupy lateral sides of the frame **101**.

[0085] Different arrangements of cooperating accessories are illustrated in the embodiments of the Figures. In FIG. **1**, the system **100** is illustrated with the cargo rack **225** installed on the frame **101**. In FIGS. **2** to **4**, the system **100** is illustrated with the cargo rack **225** and one toolbox **200** (partially cut-away for illustrative purposes) installed on the frame **101**.

[0086] In FIGS. **8** to **12**, the system **100** is illustrated with the cargo rack **225**, two sets of ladder racks **270** (including one with a lifting system **271**) shown holding ladders **50**, two toolboxes **200**, and the cab rack extension **230** installed simultaneously on the frame **101**. In FIG. **13**, the system **100** is illustrated with one rack member **221**, two toolboxes **200**, the cab rack extension **230**, the canopy **250**, and the rear canopy door **260** installed simultaneously on the frame **101**. In FIGS. **14** and **15**, the system **100** is illustrated with one rack member **221**, one toolbox **200**, the cab rack extension **230**, the canopy **250**, the rear canopy door **260**, and one side door **280** installed simultaneously on the frame **101**. The side door **280** permits access to an interior of the pickup bed through a lateral side of the system **100**.

[0087] With additional reference to FIGS. **16** and **17**, the toolboxes **200** will be described in more detail. As is illustrated in FIGS. **8** to **15**, each toolbox **200** occupies an entirety of the lateral opening in the frame **101** formed between the forward and rearward disposed rising frame members **130**.

[0088] The toolbox **200** includes a toolbox body **205** defining a cavity **208** configured and shaped to receive objects therein. Interior walls of the body **205** extend generally vertically (inner most wall) and horizontally (top and bottom walls) when installed in the frame **101** (see for instance FIGS. **9** and **16**). A top side of the toolbox body **205** is thus generally flat, permitting selective installation of both the toolbox **200** and the canopy **250**, as mentioned above. In the illustrated embodiment, the body **205** has shelves and organizers disposed therein, but the particular interior arrangement is not meant to be especially limited. The toolbox **200** also includes a toolbox door **210** pivotally connected to the toolbox body **205**, the toolbox door **210** selectively closing the cavity **208**. When the toolbox **200** is installed in the frame **101**, the toolbox door **210** extends generally parallel to the rising members **130** (when the door **210** is in a closed position).

[0089] When the toolbox **200** is connected to the frame **101**, the toolbox body **205** closes the lateral side opening of the frame **101**. The toolbox door **210** opens to an exterior of the frame **101**. Users of the system **100** can thus access items stored in the cavity **208** from an exterior of the vehicle **10**.

[0090] To aid in maintaining the integrity of items inside the toolbox body **205**, the toolbox door **210** includes a sealing gasket **212** disposed on an interior side of the toolbox door **210**. The sealing gasket **212** provides a sealing closure of the toolbox door **210**, impeding the entry of rain or the elements into the toolbox body **205**. To further aid in protecting items inside the toolbox **200** from the elements, the toolbox body **205** forms a gutter arrangement **207** therein. The gutters **207** provide a flow path for rain or fluids incident on the toolbox **200**, away from the cavity **208**.

[0091] By the present technology, the toolbox door **210** includes a latch **215** defined in a bottom side of the toolbox door **210** and hinges (not shown) connecting an upper edge of the toolbox door **210**. A handle **216** of the toolbox door **210** for selectively opening the latch **215** is also disposed on the bottom side of the toolbox door **210**. By positioning the latch **215** and the handle **216** on the bottom side of the toolbox door **210**, a greater area on the lateral side of the door **210** is available for marking, aesthetic additions, advertising, and the like. The latch **215** and the handle **216** are also at least partially sheltered from the elements, which can aid in reducing freezing or blocking thereof. For at least some users, the handle **216** location is more ergonomic than a more traditional lower-door face positioning for a handle.

[0092] The toolbox **200** further includes a composite end cap **218** connected to a rear side of the toolbox body **205** (see FIG. **9**). The composite end cap **218** is fastened or adhesively affixed to the toolbox body **205**. The composite end cap **218** provides a surface for selectively fastening objects to the toolbox body **205**. The end cap **218** may be drilled, nailed, and otherwise modified, without drilling or otherwise making holes in the toolbox body **205**. In at least some embodiments, both ends of the toolbox **200** could have a composite end cap **218**. In such embodiments, the toolbox **200** could be connected to the right or left side of the frame **101**.

[0093] Modifications and improvements to the above-described embodiments of the present technology may become apparent to those skilled in the art. The foregoing description is intended to be exemplary rather than limiting. The scope of the present technology is therefore intended to be limited solely by the scope of the appended claims.

Claims

1. An accessory system for a vehicle having a pickup bed, the system comprising: a frame configured for selectively connecting to a rear portion of the vehicle, the frame being connected to and extending over the pickup bed when the frame is installed on the vehicle, the frame being arranged and configured to selectively receive a plurality of cooperating accessories, each accessory of the plurality of cooperating accessories being configured to connect to the frame individually and simultaneously with other accessories of the plurality of cooperating accessories.
2. The accessory system of claim 1, further comprising at least one accessory of the plurality of cooperating accessories.

3. The accessory system of claim 1, wherein the plurality of cooperating accessories comprises at least one of: at least one toolbox; at least one top rack member; a cargo rack formed from a plurality of rack rigid members; a cab rack extension; a canopy; a rear door; a ladder rack; at least one mounting hook; at least one side door; and a cab barrier.
4. The accessory system of claim 3, wherein: the at least one toolbox comprises: a toolbox body defining a cavity configured and shaped to receive objects therein, and a toolbox door pivotally connected to the toolbox body, the toolbox door selectively closing the cavity; and when the at least one toolbox is selectively connected to the frame: the toolbox body closes a lateral side opening of the frame, and the toolbox door opens to an exterior of the frame.
5. The accessory system of claim 1, wherein the frame comprises: a first base frame member configured to selectively connect to a first side wall of the pickup bed; a second base frame member configured to selectively connect to a second side wall of the pickup bed; two first rising frame members connected to the first base frame member, a lower end portion of each first rising frame member being connected to the first base frame member; two second rising frame members connected to the second base frame member, a lower end portion of each second rising frame member being connected to the first base frame member; a first top frame member connected to upper end portions of the two first rising frame members; a second top frame member connected to upper end portions of the two second rising frame members; and a third top frame member extending between and connected to the first top frame member and the second top frame member.
6. The accessory system of claim 5, wherein the frame comprises a fourth top frame member extending between and connected to the first top frame member and the second top frame member, the fourth top frame member being disposed rearward of the third top frame member.
7. The accessory system of claim 5, wherein: the first top frame member extends parallel to the first base frame member; and the second top frame member extends parallel to the second base frame member.
8. The accessory system of claim 5, wherein: each of the first rising frame members extends upward and inward from the first base frame member to the first top frame member; and each of the second rising frame members extends upward and inward from the second base frame member to the second top frame member.
9. The accessory system of claim 5, wherein the third top frame member comprises: a rigid member; and at least one extension member moveably connected to the rigid member, the at least one extension member being selectively slidable relative to the rigid member permitting an adjustment of a width of the frame.
10. The accessory system of claim 5, wherein each of the first rising frame members and each of the second rising frame members has two slots defined therein, the two slots extending from the lower end portion to the upper end portion.
11. The accessory system of claim 10, wherein the two slots of each of the first rising frame members and each of the second rising frame members is shaped to selectively secure at least one fastener therein, each slot having an opening narrower than a largest width of the slot.
12. The accessory system of claim 5, wherein each of the first top frame member and the second top frame member comprises: a top frame member body having four sides extending from a rear of the frame to a front portion of the frame, the top frame member body having at least one slot defined in each of the four sides.
13. The accessory system of claim 12, wherein each of the four slots of the first top frame member and the second top frame member is shaped to selectively secure at least one fastener therein, each slot having an opening narrower than a maximum width of the slot.
14. The accessory system of claim 5, wherein the first base frame member, the second base frame member, the two first rising frame members, the two second rising frame members, the first top frame member, the second top frame member, and the third top frame member are formed from extruded aluminum.

- 15.** The accessory system of claim 5, wherein each of the first base frame member and the second base frame member further comprises a plurality of clamps for selectively connecting the frame to side walls of the pickup bed.
- 16.** A toolbox for an accessory system, the toolbox comprising: a toolbox body defining a cavity configured and shaped to receive objects therein, the toolbox body being shaped and arranged to be received in an accessory system frame; and a toolbox door pivotally connected to the toolbox body, the toolbox door selectively closing the cavity, the toolbox door including: a latch defined in a bottom side of the toolbox door, and a handle for selectively opening the latch, the handle being disposed on the bottom side of the toolbox door.
- 17.** The toolbox of claim 16, wherein the toolbox door further comprises a sealing gasket disposed on an interior side of the toolbox door.
- 18.** The toolbox of claim 16, further comprising at least one hinge pivotally connecting the toolbox door to the toolbox body via an upper edge of the toolbox door.
- 19.** The toolbox of claim 16, wherein the toolbox body forms a gutter arrangement.
- 20.** The toolbox of claim 16, further comprising at least one composite end cap for selectively fastening objects to the toolbox body.
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